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RESEARCH ARTICLE

Simultaneous Detection of Drug-resistant Mutations in *Mycobacterium tuberculosis* and Determining their Role through *in silico* Docking

Somanna Ajjamada Nachappa¹, Sumana M. Neelambike², Ahmad Sarikhani¹ and Nallur B. Ramachandra^{1,*}

¹Genetics and Genomics Lab, Department of Studies in Genetics and Genomics, University of Mysore, Mysuru, Karnataka, India; ²Department of Microbiology, JSS Medical College and Hospital, Mysuru, Karnataka, India

Abstract: A molecular method for diagnosis of drug-resistant Tuberculosis is Multiplex allele-specific PCR (MAS-PCR), which is more time-efficient, and its accuracy is studied using DNA sequencing. Also, understanding the role of mutations, when translated to protein, in causing resistance helps in better drug designing.

Aim: To study MAS-PCR in the detection of drug resistance in comparison to DNA sequencing in *Mycobacterium tuberculosis*, and understand the mechanism of interaction of drugs with mutant proteins.

Methods: MAS-PCR was used for the detection of drug-resistant mutations and validation was

done through DNA sequencing. MAS-PCR targeted four genes, *iniA* for the drug Ethambutol, *rpsL* and *rrs* for Streptomycin, and *gyrA* for Fluoroquinolone resistance, respectively. Further, the sequence data was analysed and modeled for *in silico* docking to study the effect on the interaction of the anti-TB drug molecule with the target protein.

Results: We identified drug-resistant mutations in four out of 95 isolates with one of them carrying a mutation at codon *iniA501*, two at *gyrA94*, and one for both *iniA501* and *gyrA94* using MAS-PCR. DNA sequencing confirmed drug-resistant mutations in only two isolates, whereas two others had mutation adjacent to the target allele. Drug-protein docking showed Estimated Free Energy of Binding to be higher for Fluoroquinolone binding with GyrA D94V mutant. Both wild and mutant *IniA* interact with EMB but had no significant effect on binding energy.

Conclusion: DNA sequencing-based drug resistance detection of TB is more accurate than MAS-PCR. Understanding the role of mutations in influencing the drug-protein interaction will help in designing effective drug alternatives.

Keywords: Tuberculosis, resistance, ethambutol, fluoroquinolone, docking.

1. INTRODUCTION

The infectious disease Tuberculosis (TB) is caused by the bacilli *Mycobacterium tuberculosis*. The disease is airborne and primarily infects the lungs in humans. The standard treatment against active TB involves a combination of first-line drugs, i.e., Rifampicin, Isoniazid, Pyrazinamide and Ethambutol, administered for six to nine months. Many times, patients do not strictly follow the drug regime, which not only causes treatment failure but the possibility of developing drug resistance. *Mycobacterium tuberculosis* acquires drug resistance through random and single-step mutations occurring in large populations [1]. Fluoroquinolone is a class

of second-line antibiotics administered to patients with resistance towards first-line drugs.

From the viewpoint of preventable and curable diseases, Tuberculosis remains the leading cause of death in the world. One-fourth of the globally incident TB cases annually occur in India. In 2017, out of the estimated global annual incidence of 10 million TB cases, 2.7 million were estimated to have occurred in India [2, 3]. However, among the cases reported in 2016 alone, approximately 36 thousand people needed TB retreatment because of initial treatment relapse, failure or loss to follow up [4]. In 2017, the Karnataka state had 81,187 patients registered for TB treatment, of which a little over 10 thousand were previously treated. Among the 27,397 newly registered TB cases for the year 2016 in Karnataka, 2 percent showed treatment failure, and 5 percent lost to follow-up [4]. In India, the incidence of multi-drug resistant TB (MDR-TB) in 2017 is estimated to be ~135,000 [3].

* Address correspondence to this author at the Genetics & Genomics, University of Mysore, Mysuru, Karnataka, India-570006; Tel: 91-821 2419781/888; E-mail: nbuom@gmail.com

The emergence of drug resistance in *Mycobacterium tuberculosis* poses a severe threat to global TB control. Drug-resistant TB treatment requires second-line drugs, which are less effective, more expensive, and more toxic [5].

Ethambutol (EMB) resistance mutations are primarily found in isoniazid inducible protein *IniA* (*iniA*) gene, which plays a role in efflux transmembrane transport activity. The substitution of serine to tryptophan at codon 501 (S501W) comprises the majority of *iniA*-associated mutations and confers high-level EMB resistance [6, 7]. Resistance to Streptomycin (SM), another first-line drug, is primarily caused by mutations in *rpsL* and *rrs* gene. The *rpsL* gene codes for the 30S ribosomal protein S12, and *rrs* is a non-protein coding gene, transcribing into 16S rRNA. They are involved in protein synthesis and elongation, a process which is targeted by SM [8-10]. The *gyrA* gene codes for the DNA gyrase subunit A, which is a type II topoisomerase that negatively supercoils closed circular double-stranded DNA and carries most of the known mutations responsible for resistance towards Fluoroquinolone (FLQ). The high confidence mutation in *gyrA* gene is located in codon 94 that codes for amino acid aspartic acid [11, 12].

As drug-resistant TB is difficult to treat, rapid diagnosis and prevention of transmission within the community are imperative. The gold standard to diagnose resistant TB is culture-based drug susceptibility testing (DST), but this significantly delays diagnosis due to the slow replication time of *M. tuberculosis*. Study of the genetic mechanisms underlying *M. tuberculosis* drug resistance and advances in technology have promoted the development of rapid molecular diagnostics. Simultaneous detection of the most common resistance-associated genetic mutations at relatively reduced cost and need for technical expertise is possible using Multiplex allele-specific PCR (MAS-PCR) [13]. By following up MAS-PCR with systematic identification of the genes through DNA sequencing and bioinformatics analysis, we evaluated the ability of the MAS-PCR to detect first- and second-line drug-resistant mutations. Further, looking into the effect of these mutations on Mycobacterial protein interaction with the drug molecule offers excellent potential in terms of drug target discovery and development of new antibacterial agents [14].

2. MATERIALS AND METHODS

2.1. Ethics Statement

The following study was carried out after obtaining approval from the Institutional Human Ethical Committee of the University of Mysore, Mysuru.

2.2. *M. tuberculosis* DNA Isolates

Genomic DNA samples used for the study had been extracted from *M. tuberculosis* cultures of clinical isolates and stored at -20 °C. The samples had been previously obtained in the form of sputum from unrelated patients positive for Ziehl-Neelsen (ZN) staining, but not classified as carrying resistant TB. These samples were used for culturing *Mycobac-*

terium on solid Lowenstein-Jensen media. The bacterial colonies were scooped and DNA was isolated using the QIAamp DNA Mini Kit (Qiagen, Germany).

2.3. Multiplex Allele-Specific Polymerase Chain Reaction (MAS-PCR)

A total of 95 *M. tuberculosis* isolates were analysed using MAS-PCR for drug-resistant mutations on four genes, *iniA*, *rpsL*, *rrs* and *gyrA*. The *M. tuberculosis* isolate DNA was used in the MAS-PCR assay to target codon 501 of the *iniA* gene associated with Ethambutol drug resistance [7], the codon 43 of the *rpsL* gene and the *rrs* gene 514bp associated with Streptomycin drug resistance [8, 9], and *gyrA* gene codon 94 involved in Fluoroquinolone resistance [11]. Allele-specific primers for the respective gene were designed using the NCBI Primer-BLAST tool [15]. The 3' end of each allele-specific primer was paired with the wild-type allele of the respective codons where the target point mutations have been found. The allele-specific primers for the four loci (Table 1) were combined into a single PCR reaction mix to detect mutations based on the presence and size of the amplification product (Table 1). The primers target wild-type allele, thus the presence of mutation does not allow allele-specific amplification and the band corresponding to that amplicon size is absent in the gel.

The reaction mix for MAS-PCR contained the eight primers targeting the four loci as mentioned in Table 1, Genei PCR Master Mix 2X (Merck, India) and template *M. tuberculosis* DNA in a 50 µl reaction made up with molecular grade water. The thermocycling parameters included an initial denaturing step at 96 °C for 3 min, 25 cycles involving denaturation at 95 °C for 50 s, annealing at 64 °C for 40 s and extension at 72 °C for 1 min, and a final extension at 72 °C for 7 min. The banding pattern of the PCR products was examined by 2.5% agarose gel electrophoresis in 1X Tris-EDTA buffer.

2.4. DNA Sequencing

M. tuberculosis isolates DNA analysed with MAS-PCR were sequenced to confirm the presence of the MAS-PCR detected mutations. Primers were designed using the NCBI Primer-BLAST tool [15] to flank the region containing the mutations, amplified with PCR and sequenced by the Sanger method. The primers used to amplify the region are listed in Table 2 for each gene. The nucleotide sequence obtained for the respective genes was aligned against the reference gene sequence of *M. tuberculosis* H37Rv strain using EMBOSS Needle [16]. The sequence data was used to confirm the presence of the mutations detected using MAS-PCR.

2.5. Automated Molecular Docking

The mutated sequence of the *iniA* and *gyrA* genes was used for studying the interaction of their proteins with the respective anti-mycobacterial drug Ethambutol and Fluoroquinolone *in silico*. Ofloxacin (OFLX) was selected as the ligand for the docking study, which is one of the most potent drugs in the FLQ class [17, 18]. The *IniA* (reference se-

Table 1. Primers for MAS-PCR to detect SM, FLQ and EMB resistance mutations in *M. tuberculosis*.

Target Genes (Amino Acid)	Drug Resistance	Allele-specific Primers (5' – 3')*	Paired Primers	PCR Products
<i>iniA501</i> (Ser/TCG)	Ethambutol	GTCATGATTGGCATGCTGTC	GGTGATCTCTTCGGCGATCT	264
<i>rpsL43</i> (Lys/AAG)	Streptomycin	GTGTACACCACCACTCCGAA	ACCGCGGATGATCTTGTAG	192
<i>rrs514</i> (-)	Streptomycin	CGGCCAACTACGTGCCAGCA	TTTACGGCGTGGACTACCAG	313
<i>gyrA94</i> (Asp/GAC)	Fluoroquinolone	CCATGCGCACCAGGCTGT	CAGCGCAGCTACATCGACTA	226

*Nucleotides highlighted in bold are complementary to the allele being investigated for mutation

Table 2. PCR primers for amplifying and DNA sequencing of drug resistance genes.

Target Genes	Forward Primers (5' – 3')	Reverse Primers (5' – 3')	Annealing Temp.	PCR Products
<i>iniA501</i>	AACGACGTCGAGAATGCGAT	GCCGATCCCGTGATTGTTT	60 C	463 bp
<i>gyrA94</i>	ATGACAGACACGACGTTGCC	TCGACTGTCTCCTCGTCGAT	60 C	452 bp

quence: NC_000962.3) and GyrA (reference sequence: NC_000962.3) protein wild-type reference amino acid sequences were obtained from the NCBI database. The respective mutated protein sequences were generated using the ExPASy Translate tool [19] from the DNA data obtained through Sanger sequencing.

SWISS-MODEL [20-23] was used to generate the three-dimensional protein structure of the IniA and GyrA proteins to be used in the docking study. For *M. tuberculosis* IniA, the 3D structure of *Nostoc punctiforme* Dynamin-like protein BDLP at 3.0Å resolution (PDB ID: 2J69) was used as a template for creating the wild-type and mutant protein model [24]. Crystal structure of the topoisomerase II from *M. tuberculosis* at 2.55Å resolution (PDB ID: 5BTA) was used as a template to generate the GyrA monomer [25]. The drug molecules, Ethambutol 3D structure (CID: 14052) [26] and Ofloxacin 3D structure (CID: 4583) [27] for the docking study with IniA and GyrA proteins, respectively were obtained from the PubChem database.

The *in silico* docking was carried out using the program DockingServer [28]. For the IniA protein, the search box values of 20, 20, 20 in the X, Y, Z dimensions were selected around the codon 501 where the mutation was observed and docking program was run using Ethambutol 3D structure as a ligand. In the case of GyrA, the search box values were 20, 20, 20 in the X, Y, Z dimensions centered on the codon 94 to include the active site. The 3D structure of the drug Ofloxacin was used as a ligand for docking against the receptor GyrA.

3. RESULTS

3.1. MAS-PCR

95 DNA samples extracted from *M. tuberculosis* clinical isolates were analysed for drug resistance mutations using MAS-PCR for four alleles on as many genes- *iniA501*, *rpsL43*, *rrs514* and *gyrA94*. As visualised by agarose gel elec-

trophoresis of the PCR products, the MAS-PCR generated distinct banding patterns for different mutation profiles (Fig. 1). Among the 95 DNA isolates, four showed mutations conferring drug-resistance. Among them, one had mono-resistant mutation at *iniA501*, two at *gyrA94*, and one isolate showed mutations at both *iniA501* and *gyrA94* (Table 3). The remaining 91 samples showed no mutations at the four locations.

3.2. DNA Sequencing

DNA sequencing of the *iniA501* and *gyrA94* containing regions of respective genes was carried out for all the four *M. tuberculosis* isolates, which showed mutations with MAS-PCR. The sequencing of 463 bp region of *iniA* containing codon 501 revealed an S501W point mutation in one of the two MAS-PCR detected isolates, but this isolate also carried a point mutation leading to the change of histidine to glutamine at codon 481 (H481Q). Sequence data of the 452 bp region of the *gyrA* gene covering the codon 94 revealed D94V mutation in one sample but showed serine to threonine (S95T) point mutations at codon 95 in two different isolates (Table 3).

3.3. Automated Molecular Docking

The Ethambutol and Ofloxacin drug molecule structures obtained from PubChem were docked into the manually defined binding site of the wild-type and mutant *Mycobacterium tuberculosis* IniA and GyrA protein models, respectively, with DockingServer (Fig. 2). The program provides information on the protein side chains interacting with the drug ligand and data on the Estimated Free Energy of Binding (ΔG) for each docking interaction. The amino acid serine at position 501 in wild-type shows contact interaction with Ethambutol, and tryptophan in the mutant IniA protein shows the same along with weak hydrophobic interaction with Ethambutol. Further mutant IniA carrying the additional mutation H481Q with S501W was modeled, and this protein, when docked with Ethambutol, showed no change in

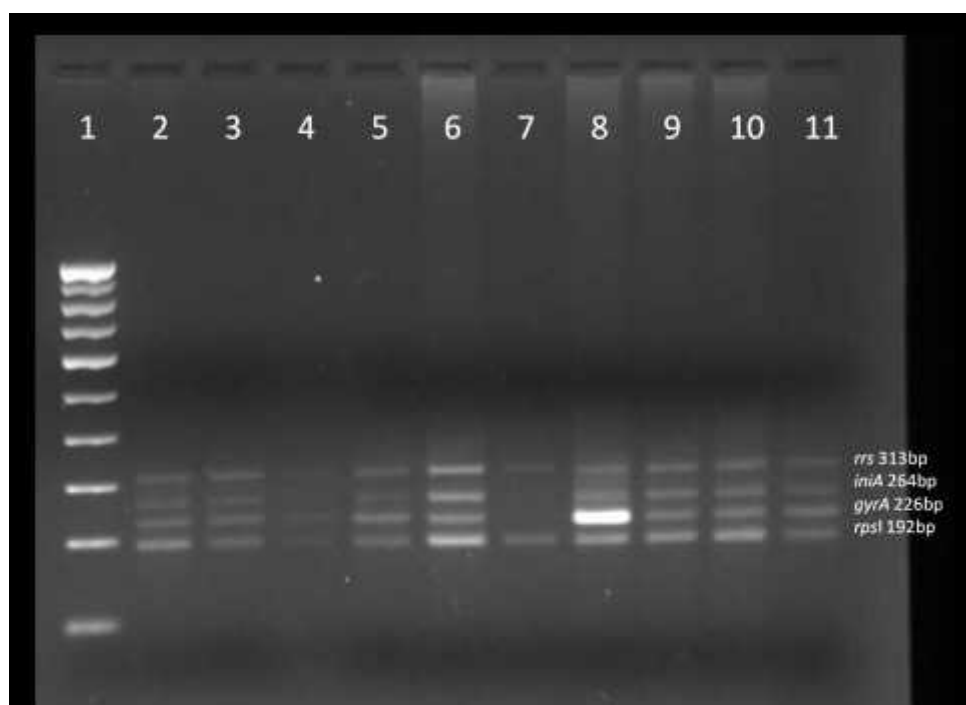


Fig. (1). Drug resistance profile of *M. tuberculosis* isolates by MAS-PCR in 2.5% agarose gel. Band sizes correspond to mutations of genes, as mentioned in Table 1. The first lane represents the 100-bp step-up DNA ladder. Lane 2, 3, 4, 5, 6, 8, 9, 10, 11 represent isolates with no mutations. Lane 7 represents mutation in *iniA* codon 501 and *gyrA* codon 94. The respective size of amplicons mentioned on the right-hand side.

Table 3. Comparison of the results of MAS-PCR and DNA Sequencing.

Isolate Code*	MAS-PCR		DNA Sequencing				
	EMB Resistance	FLQ Resistance	<i>iniA</i> Gene		<i>gyrA</i> Gene		Drug resistance
			Sequence Change	Amino-acid Change	Sequence Change	Amino-acid Change	
A	<i>iniA</i> 501	-	-	-	-	-	-
B	<i>iniA</i> 501	<i>gyrA</i> 94	<i>iniA</i> T1443G+ <i>iniA</i> C1502G	<i>iniA</i> H481Q+ <i>iniA</i> S501W	<i>gyrA</i> G284C	<i>gyrA</i> S95T	Ethambutol
C	-	<i>gyrA</i> 94	-	-	<i>gyrA</i> A281T	<i>gyrA</i> D94V	Fluoroquinolone
D	-	<i>gyrA</i> 94	-	-	<i>gyrA</i> G284C	<i>gyrA</i> S95T	-

*Isolate code for representation purpose only

Gene mutations highlighted in **bold** are associated with respective drug resistance

interaction as compared to IniA protein carrying the sole S501W mutation. In the case of GyrA protein, the aspartic acid at codon 94 in the wild-type forms part of the interacting side chains with the Ofloxacin molecule but shows no direct bond formation with the drug molecule. Valine at the same position in mutant GyrA shows similar interaction as aspartic acid with the ligand. Fig. (3) highlights all the protein residues interacting with the respective drug molecule. The ΔG for the wild, S501W and S501W+H481Q mutant IniA-EMB interaction are -4.58 kcal/mol, -3.86 kcal/mol, and -3.67 kcal/mol, respectively. For the interaction between wild and mutant GyrA with Ofloxacin, the ΔG values are -5.89 kcal/mol and +12.67 kcal/mol, respectively (Table 4).

4. DISCUSSION

M. tuberculosis genomic DNA were obtained from 95 TB patients with no prior diagnosis of drug resistance to study four candidate genes for the presence of high confidence mutations, *iniA*501, *rpsL*43, *rrs*514 and *gyrA*94, to determine as either being resistant or sensitive to one or more of the three drugs considered for this study, i.e., Ethambutol, Streptomycin and Fluoroquinolone. The mutations were targeted as they showed the highest frequency compared to other drug-resistant mutations in the corresponding resistant *M. tuberculosis* strains [6-9, 29-31]. The drugs were selected to look for the presence of drug-resistant mutations towards the less potent First-line drugs Ethambutol and Strepto-

mycin, and also a class of second-line drug Fluoroquinolone, which are not usually considered while diagnosing drug resistance in clinical patients.

Most widely used methods to detect drug-resistant mutations include PCR and DNA sequencing [31]. We used MAS-PCR to initially screen all the samples for EMB, SM and FLQ resistance by targeting mutations at four different locations (Table 1), as it allows for faster simultaneous detection of mutations at multiple loci thus increasing the efficiency substantially and reducing significantly the detection cost [5, 13, 32]. Four isolates showed single nucleotide mutations at one or more of the four loci being targeted, conferring drug-resistance across the isolates (Table 3). MAS-PCR predicted two of the isolates as resistant to EMB, one to FLQ, and one as resistant to both EMB and FLQ. The remaining 91 samples showed no mutations at the four loci suggesting susceptibility to all the three drugs, *i.e.*, EMB, SM and FLQ. The low number of resistant isolates detected are in concordance with the actual number of cases observed in the random cohort carrying resistant strains, in agreement with the WHO report, which states that about 2.2 percent of the newly reported cases in India are drug-resistant [4, 33].

DNA sequencing of the *iniA501* and *gyrA94* containing regions of respective genes for the four isolates confirmed only two isolates, *i.e.*, Isolate B and C as drug-resistant, where both were mono drug-resistant (Table 3). This valida-

tion step allowed us to identify the false-positive isolates (Isolate A and D) as a result of MAS-PCR. Isolate B earlier classified as resistant towards EMB and FLQ, showed drug-resistant mutation for only EMB (*iniA501*), whereas FLQ associated *gyrA* gene carried the mutation causing the change S95T, which is not known to cause resistance towards fluoroquinolones [34].

The *gyrA* point mutation at nucleotide 284G → C causing the change S95T causes false detection with MAS-PCR as it lies close to the 3' end of the allele-specific primer for detecting D94V mutation. The presence of *gyrA* S95T mutation though not associated with drug resistance, can give false-positive results with PCR and lead to unwarranted drug treatment until resistance is confirmed with phenotypic DST. A better primer design may be able to circumvent this problem, but the allele-specific primer designed by us for the nucleotide 281 (codon 94) on the complementary strand was not robust enough as the one used in this study (Table 1) and failed to produce amplification product. Misdiagnosis as drug-resistant can lead to prescribing treatment with a different cocktail of anti-TB drugs, which could be less effective, more expensive and toxic for the patient [35, 36]. Performing DNA sequencing on every occasion to validate MAS-PCR might not be feasible and thus, we believe sequencing alone would be sufficient and a better option as it is more reliable and will save additional resources.

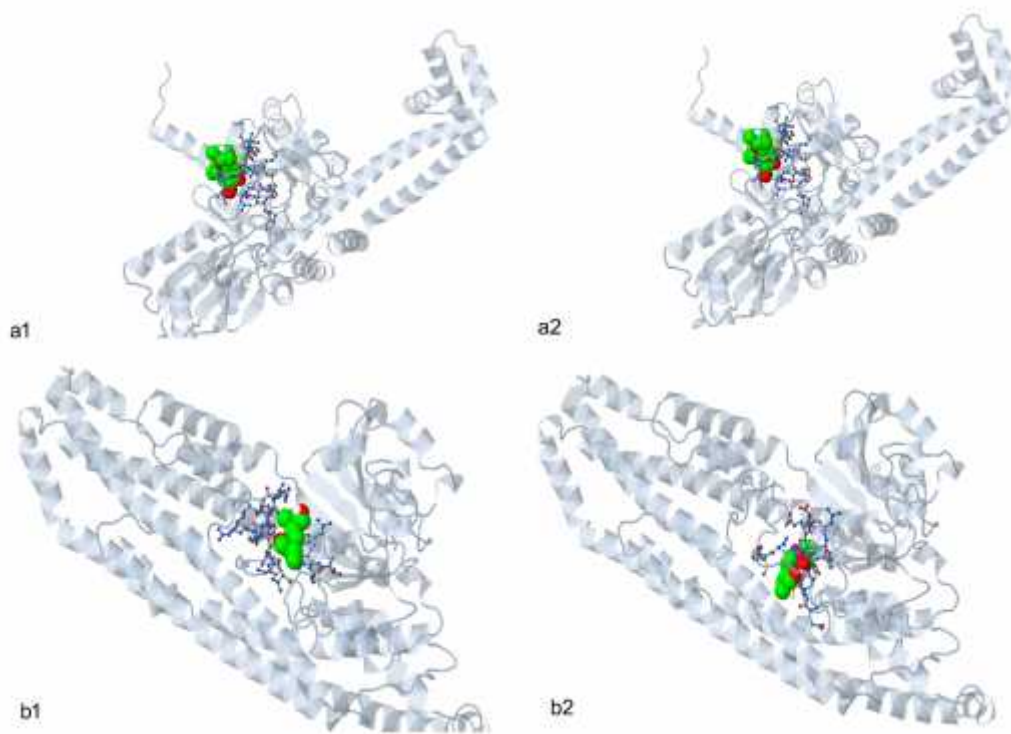


Fig. (2). Interaction between, **a1**: GyrA wild protein and Ofloxacin molecule, **a2**: GyrA D94V mutant protein- Ofloxacin, **b1**: IniA wild protein-Ethambutol, and **b2**: IniA S501W mutant protein- Ethambutol. Grey colour animated structure represents the protein, Green spheres form the ligand, and Blue-Red balls/sticks are the interacting side chains of the protein.

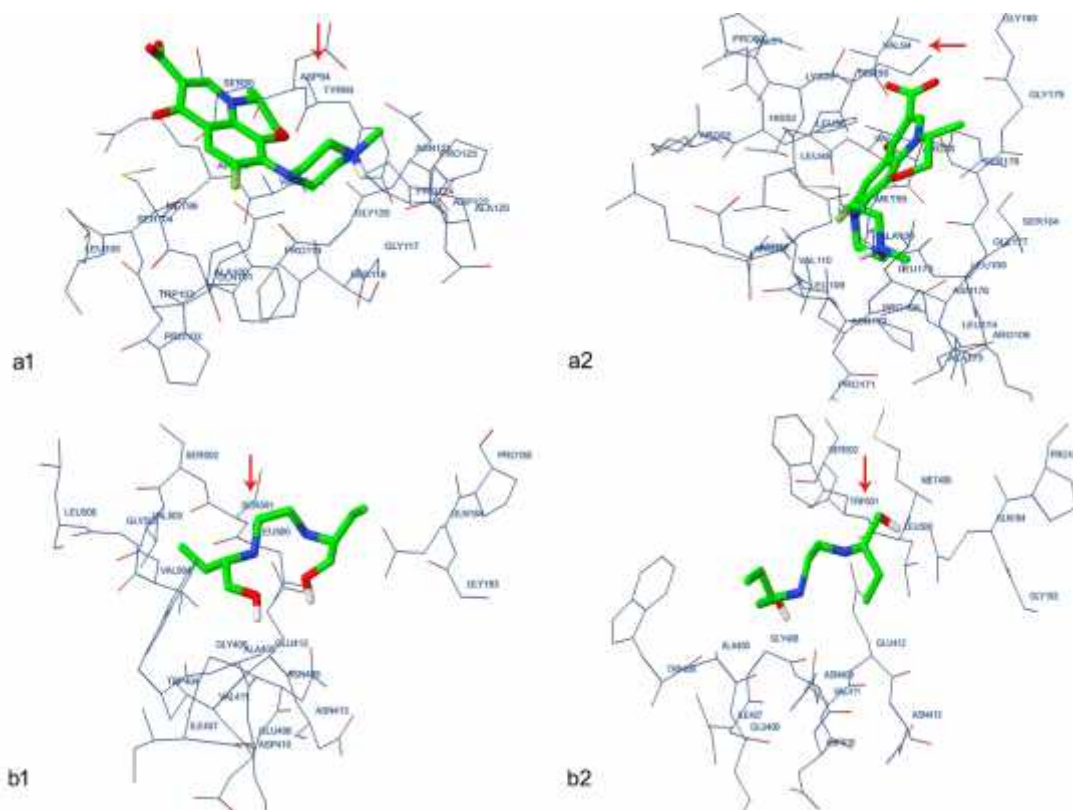


Fig. (3). Amino acid side chains interacting with respective drug molecule, **a1**: GyrA wild protein and Ofloxacin molecule, **a2**: GyrA D94V mutant protein- Ofloxacin, **b1**: IniA wild protein-Ethambutol, and **b2**: IniA S501W mutant protein- Ethambutol. Green molecules are the ligands, and the thin Blue sticks represent the amino acids interacting with the ligand. The red arrows point towards the wild and respective mutant amino acid.

Table 4. Comparison of the estimated free energy of binding between the wild and mutant IniA and GyrA proteins.

Protein	Modeling Template	Ligand	Free Energy of Binding ΔG (kcal/mol)	Remarks
IniA wild	2J69 (<i>E. coli</i>)	Ethambutol	-4.58	-ve ΔG implies favourable thermodynamic environment for binding inside the cell & high probability of drug-protein interaction at the binding site
IniA S501W	2J69 (<i>E. coli</i>)	Ethambutol	-3.86	
IniA S501W+H481Q	2J69 (<i>E. coli</i>)	Ethambutol	-3.67	
GyrA wild	5BTA (<i>M. tuberculosis</i>)	Ofloxacin	-5.89	High +ve ΔG indicates energy-demanding molecular interaction between the drug & target, reducing chances of drug binding
GyrA D94V	5BTA (<i>M. tuberculosis</i>)	Ofloxacin	+12.67	

-ve: negative; +ve: positive

Among the isolates tested by MAS-PCR and validated through DNA sequencing, Isolate B showed an interesting additional mutation at the codon 481 of the *iniA* gene (Table 3). Since we identified this non-synonymous mutation in a region adjacent to the known resistance causing *iniA501* codon mutation, we wanted to understand whether this additional mutation plays any role in augmenting the role of the primary drug-resistant mutation. Thus, to understand the role of these sequencing-validated mutations, we looked into the molecular interaction between the anti-TB drug and the mutant protein to understand the effect of the mutation on the dynamics of drug-protein binding. The EMB molecule was

docked into the IniA protein and OFLX with the GyrA protein, as these drugs target the respective protein molecule inside the bacterial cell and are essential for the effective drug response against TB.

The non-synonymous *iniA* gene mutation S501W did not interfere in the normal interaction between EMB and IniA. Though tryptophan shows weak hydrophobic interaction with Ethambutol, this led to no significant change in the estimated free energy of binding (ΔG) with the EMB molecule compared to the wild-type IniA protein. Further, IniA with both non-synonymous changes, i.e., H481Q + S501W did not deviate from the ΔG observed for wild and S501W mu-

tant protein. The results show that the non-synonymous mutation at codon 501 in IniA though associated with EMB resistance, and the presence of an additional mutation at *iniA481* may not solely play a significant role in the binding of the drug at the active site. The occurrence of the additional H481Q mutation does not alter protein-drug binding interactions, which might be due to similar polar property of both histidine and mutant glutamine. The previous study by Hazbon *et al.* on the H481Q mutation suggests that the mutation is not linked to selection pressure and not likely to confer drug resistance or fitness to *M. tuberculosis* [37].

The mutant valine at codon 94 in GyrA protein led to an increase in ΔG from -5.89 kcal/mol in wild-type to +12.67 kcal/mol for OFLX binding. Aspartic acid and valine at codon 94 show no direct bond formation but are in the group of interacting molecules. GyrA94 aspartic acid is predicted to anchor the bridge between FLQ and the enzyme and mediate their interaction [38]. The difference in the intrinsic property of the two amino acids might impact the molecular dynamics for the binding affinities with OFLX, as aspartic acid is hydrophilic and carries a negative charge at neutral pH, in contrast to valine which is highly hydrophobic. The increased ΔG reduces the chances of interaction between the drug and the target protein. This cannot be compensated by increasing Ofloxacin dosage without increasing toxicity, making the drug ineffective and combating the TB infection difficult.

CONCLUSION

MAS-PCR allows the simultaneous detection of mutations for large samples at reduced costs, but the results are relatively less reliable compared to DNA sequencing. We thus suggest that DNA sequencing would be a more reliable

method, which can be standardised for quicker data analysis and faster output. Operating cost makes DNA sequencing less attractive, but with increased adaptation in clinical setup, and gaining more popularity, it can become less expensive. Therefore, sequencing promises to be more precise in detecting drug-resistant strains of *Mycobacterium* and aid in the initiation of correct medication for the treatment of Tuberculosis patients. Changes in drug design are necessary for effective treatment of Tuberculosis in the case of the *Mycobacterium* strain carrying the *gyrA94* mutation which significantly affects the binding of the Fluoroquinolone class drug Ofloxacin.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

The following study was carried out after obtaining approval from the Institutional Human Ethical Committee of the University of Mysore, Mysuru (IHEC-UOM No. 32/RES/2012-13).

HUMAN AND ANIMAL RIGHTS

No animals/humans were used for studies that are the basis of this research.

CONSENT FOR PUBLICATION

Not applicable.

AVAILABILITY OF DATA AND MATERIALS

Not applicable.

FUNDING

None.

CONFLICT OF INTEREST

The author declares no conflict of interest, financial or otherwise

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